

Modern Romantic Novels — Themes × Popularity

A reader-friendly walkthrough of the pilot analysis (Stage 10, N=92)

Goal: show the measurement pipeline and the most stable thematic signals for **Reach** (how many people rate a book) vs **Quality** (how positively people rate it).

Source deck content: PRESENTATION_SLIDES_DRAFT + Stage10 step-by-step deck.

How to read this deck (pilot study)

- This is an **instrument calibration** run: the focus is on a clean, auditable pipeline (not chasing “ $p < .05$ ”).
- We track two outcomes separately: **Reach** = $\log(\# \text{ ratings})$ vs **Quality** = mean rating (with reach controlled).
- Effect sizes are emphasized (direction + uncertainty).
- Key headline: **Reach** shows clearer thematic signal than **Quality** in this pilot; **Payoff/Safety** relates most to Quality.

Dataset: 105 billionaire romance novels (92 with complete metadata), 680,822 sentences; tiers: Top (30), Middle (32), Trash (30).

Section A — Research setup

- Research question:** Which thematic patterns differentiate higher-rated romance novels from lower-rated ones?
- Topic model: BERTopic + Bayesian hyperparameter search (300+ configurations).
 - Topics → LLM labels → theory-aligned taxonomy (8 groups, 30+ categories) → indices → statistical tests.
 - Why this design? It keeps “what the model found” separate from “how we interpret it” and “what predicts ratings.”

Data element	What it is	Why it matters
Topic probabilities	Per-book probability mass across 368 topics (sums to 1)	Enables us compare books on the same scale
Taxonomy groups	Human-readable theme buckets (e.g., Relationship Trajectory, Social World)	Reduces noise; improves interpretability
Indices / macro axes	Small set of composite predictors (e.g., Explicitness, Porn/Safety)	Enables inference at small N

Section B — Picking a good topic model

Problem: topic models trade off **coherence** (topics make sense) vs **diversity** (coverage of different themes).

How we search: Bayesian optimization tries many settings and learns which hyperparameters improve the metrics.

How we decide: Pareto efficiency keeps only models where you can't improve coherence without hurting diversity (or vice versa).

B2 — Pareto frontier: coherence vs diversity (what's going on here?)

Each point = one model configuration. Points on the boundary curve are “best trade-offs.” We select a frontier model rather than guessing a single metric weight.

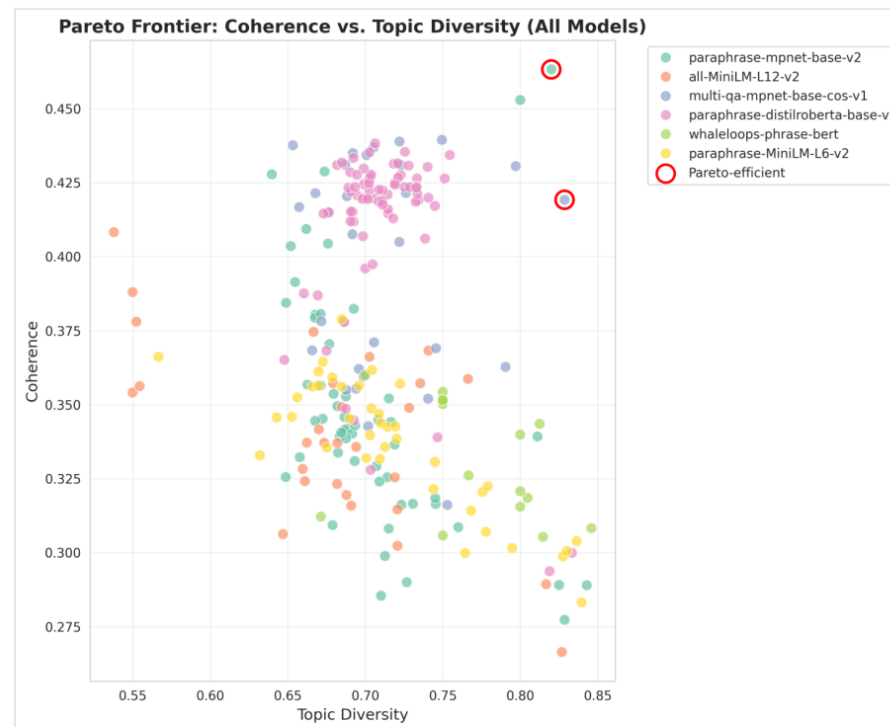


Figure: The Pareto frontier shows the optimal trade-off curve between coherence and diversity. Models on the frontier represent non-dominated solutions.

Pareto-efficient model (example)	Coherence	Diversity	Why we keep it
paraphrase-mpnet-base-v2 (Rank 1)	0.463	0.82	Best balance (highest coherence among frontier models)
paraphrase-MiniLM-L6-v2 (frontier)	0.425	0.94	Highest diversity (more fine-grained topics)

B3 — Which knobs mattered most? (hyperparameter trade-offs)

UMAP settings (how we lay out sentence embeddings) and vocabulary filtering had the strongest links to performance. Think of these as “how tightly do sentences cluster” and “how much rare vocabulary do we keep.”

Slide 9: Hyperparameter Correlation Analysis

Critical Hyperparameters Identified

High Priority Parameters (Strong, Significant Effects):

Parameter	Effect on	Correlation	p-value	Effect Size (Cohen's d)
umap__min_dist	Diversity	r = 0.794	p = 0.006	Large (d = -1.10)
vectorizer__min_df	Combined Score	r = 0.745	p = 0.014	Very Large (d = -2.56)
umap__n_components	Combined Score	r = 0.712	p = 0.021	Large (d = -1.46)

Medium Priority Parameters (Strong Effects, Marginal Significance):

Parameter	Effect on	Correlation	p-value
bertopic__min_topic_size	Coherence	r = 0.585	p = 0.075
hdbscan__min_cluster_size	Coherence	r = 0.563	p = 0.090
umap__n_neighbors	Coherence	r = -0.573	p = 0.084

Key Trade-offs Identified

1. bertopic__min_topic_size
2.  Increases coherence (r = 0.585)
3.  Decreases diversity (r = -0.591)
4. Trade-off: Filter small topics → better coherence, fewer topics
5. umap__min_dist

Parameter	What it controls (plain language)	Observed trade-off
umap__min_dist	How close points are allowed to be	↑ diversity, but can ↓ coherence
vectorizer__min_df	Drop very-rare words	Strong effect on combined score

min_topic_size	Small-topic filtering	↑ coherence, but ↓ diversity
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Section C — Micro-scenes: which topics show up more in Top vs Trash?

Method (simplified): We compare topic probabilities in Top vs Trash books using **Cliff's δ** (an effect size).

- **Cliff's δ** is the probability that a random Top-tier book has more of a topic than a random Trash-tier book, minus the reverse. ($\delta=0$ means no difference.)
- We use effect sizes + a small impact filter to avoid “tiny but significant” noise.
- These are **exploratory** micro-signals; we aggregate next to get more stable predictors.

C2 — Topic drivers heatmap + real examples

Darker cells = larger differences. This tells us which specific micro-scenes contribute most to subgroup differences.

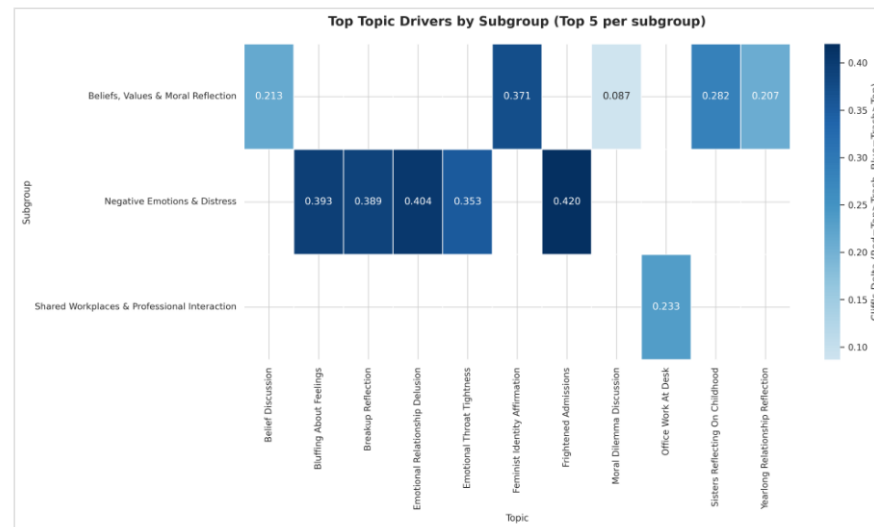


Figure: Heatmap showing which topics drive key subgroup differences, revealing the micro-scenes that contribute to tier differentiation.

Table reference: `results/stage10_correlation_analysis/01_topic_analysis/tables_csv/topic_leaderboard_tier1_high_confidence.csv`

Top-associated topic label	Cliff's δ (Top vs Trash)	Plain-language gloss
Married Couple's Affectionate Stares	0.45	Warm, emotionally credible intimacy cues
Frightened Admissions	0.42	Vulnerability + reassurance scenes
Emotional Relationship Delusion	0.40	Self-doubt / misreading signals (psych realism)

Bluffing About Feelings	0.39	Covering emotions → tension → later repair
Feminist Identity Affirmation	0.37	Values/identity reflection moments

C3 — What this means (micro-scenes → hypotheses)

- Top-tier differentiation is less about “more sex” and more about **psychological credibility**: fear admissions, self-concept, honest conflict.
- Trash-tier differentiation leans toward **procedural/transition** scenes and higher explicitness (in this corpus).
- We use these micro-results to (1) sanity-check labeling, (2) suggest theory mechanisms, (3) choose candidates for composites—without outcome-driven cherry-picking.

Example label	What it looks like in a scene (one-sentence paraphrase)
Parked Car Makeout	The couple is parked somewhere private, kissing / touching (embodied intimacy).
Frightened Admissions	A character admits fears/insecurities in a quiet moment; the partner responds warmly.

Section D — From topics to a taxonomy (and why it helps)

To reduce noise, we group 368 topics into **8 main theme buckets** and finer subgroups (30+ categories). We then compare group “probability mass” across tiers.

- Group shares sum to 1 per book (like a budget).
- This makes comparisons interpretable: “where does the book spend its thematic attention?”
- We also measure **diversity** (how spread-out a book’s thematic mixture is).

D2 — Main theme buckets: effect-size heatmap

Positive δ = Top has more of that theme; negative δ = Trash has more. The story is modest at the broad level (expected), but consistent.

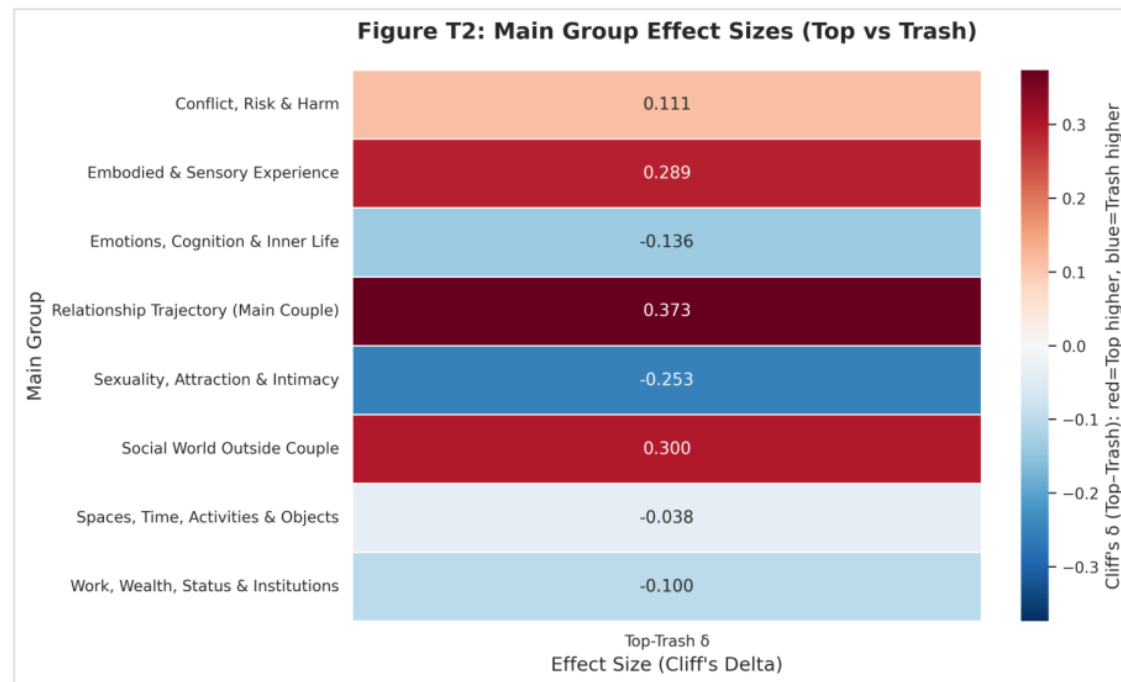


Figure: Heatmap visualization of main group differences across tiers, providing a comprehensive overview of thematic rebalancing patterns.

Main group (8 buckets)	Top vs Trash δ	Interpretation
Relationship Trajectory (Main Couple)	+0.37	More relationship-development beats
Social World Outside Couple	+0.30	More friends/work/social embedding
Embodied & Sensory Experience	+0.29	More grounded physical presence

Sexuality, Attraction & Intimacy	-0.25	More pure intimacy mass in Trash-tier
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D3 — Thematic diversity rises with tier

These metrics ask: does a book spread probability mass across many topics, or concentrate on a few? Higher-tier books show a small, monotonic increase in diversity.

Slide 15: Thematic Diversity Findings

Diversity Metrics by Tier

Monotonic trend across tiers: - **Entropy:** bad ≈ 5.33 $\rightarrow$ mid ≈ 5.43 $\rightarrow$ good ≈ 5.48 - **Effective topics:** bad ≈ 207 $\rightarrow$ mid ≈ 229 $\rightarrow$ good ≈ 240 - **Richness** (topics $> 1e-3$): bad ≈ 247 $\rightarrow$ mid ≈ 258 $\rightarrow$ good ≈ 265 - **HHI:** Lower in good (more distributed)

Statistical tests: - Entropy/effective_topics: $p \approx 0.019$ (adjusted ≈ 0.077) - Richness: $p \approx 0.015$ (adjusted ≈ 0.073)

Interpretation

Higher-tier books show greater thematic diversity — they combine more thematic ingredients rather than over-concentrating on a few motifs. This is a structural property, not tied to any single romance trope.

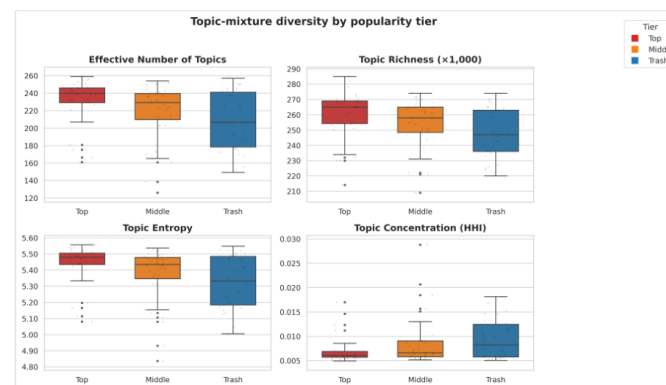


Figure: Thematic diversity metrics (entropy, effective topics, richness) showing monotonic increase across tiers.

Metric	Trash	Middle	Top	Intuition
Entropy	≈ 5.33	≈ 5.43	≈ 5.48	Higher = more even mixture
Effective topics	≈ 207	≈ 229	≈ 240	Higher = “more ingredients”

Richness (>1e-3)	≈ 247	≈ 258	≈ 265	Count of non-trivial topics
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Section E — Two channels: Reach vs Quality

Goodreads gives two different success signals. In this pilot they behave differently, so we model them separately.

- **Reach** = $\log(\# \text{ ratings})$: visibility / mass appeal.
- **Quality** = mean rating (controlling reach): how positively readers evaluate it, beyond popularity.
- Macro axes compress many indices into a few interpretable dimensions: **Status/Dominance**, **Payoff/Safety**, **Explicitness**, **Negative Affect**.

E2 — Macro axes predicting Reach (visibility)

What we're seeing: coefficients (β) show how a 1 SD increase in an axis relates to Reach. Why it matters: it separates “what sells broadly” from “what earns higher ratings.”

Slide 16: Two-Channel Analysis: Reach vs Quality

Key Finding: Reach and Quality Are Different

The analysis separates two distinct Goodreads success signals: 1. **Mass Appeal / Visibility** = `log_rating_count` (how many people rated it) 2. **Perceived Quality** = `rating_mean` (how positively readers evaluate it)

The same themes do not predict both equally well.

Mass Appeal Predictors (`log_rating_count`)

Top predictors (β with 95% CI, $P(\beta > 0)$): 1. **R2_alpha_guarding**: $\beta \approx +0.44$, CI [+0.23, +0.60], $P=1.00$ 2. **D_power_wealth_luxury__pc1**: $\beta \approx +0.37$, CI [+0.20, +0.55], $P=1.00$ 3. **A2_emotional_safety__pc1**: $\beta \approx +0.32$, CI [+0.13, +0.50], $P=0.998$ 4. **Q_repair**: $\beta \approx +0.23$, CI [+0.02, +0.41], $P=0.985$

Macro axes: - **AX_status_dominance**: $\beta \approx +0.46$, $P=1.00$ - **AX_payoff_safety**: $\beta \approx +0.33$, $P \approx 0.999$ - **AX_explicitness**: $\beta \approx -0.27$, $P \approx 0.003$ (strongly negative)

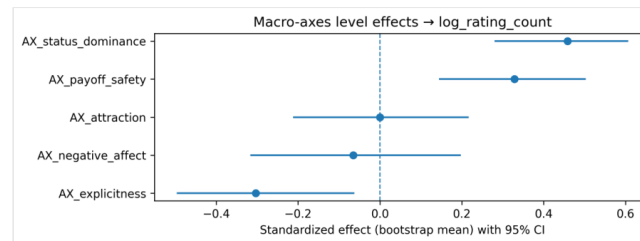


Figure: Macro-axis effects on mass appeal (`log_rating_count`). Status/dominance and payoff/safety drive reach, while explicitness is negatively associated.

Axis (macro)	β (approx.)	Direction
Status/Dominance	+0.46	More reach
Payoff/Safety	+0.33	More reach
Explicitness	-0.27	Less reach (in this corpus)

E3 — Macro axes predicting Quality (ratings beyond reach)

Controlling for Reach isolates what readers reward *after* a book is already visible. In this pilot, **Payoff/Safety** (care, repair, emotional safety) is the clearest positive signal.

Slide 17: Perceived Quality Predictors

Quality Beyond Reach (`rating_mean` , controlling for `log_rating_count`)

Top predictors: 1. `R1_protective_caretaking`: $\beta \approx +0.22$, CI [+0.05, +0.36], $P=0.995$ 2. `A2_emotional_safety__pc1`: $\beta \approx +0.15$, $P=0.95$

Macro axes: - `AX_payoff_safety`: $\beta \approx +0.21$, $P=0.974$ - `AX_explicitness`: $\beta \approx -0.15$, $P=0.095$ (tends negative) - `AX_negative_affect`: $\beta \approx -0.13$, $P=0.051$ (borderline negative)

Partial Correlations (Quality Beyond Popularity)

- `R1_protective_caretaking`: +0.245
- `A2_emotional_safety__pc1`: +0.152
- `C_explicit_eroticism`: -0.172 (negative)
- `F2_anger_frustration`: -0.121 (negative)

Interpretation

Quality beyond reach is associated with: - "Care + safety" (protective caretaking, emotional safety) - Negatively with baseline negative affect and explicit erotics

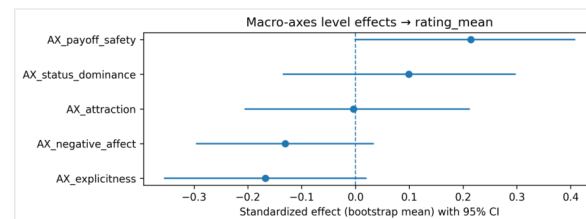


Figure: Macro-axis effects on perceived quality (`rating_mean`, controlling for reach). Care/safety themes drive quality beyond popularity.

Predictor	β / correlation	Interpretation
R1_protective_caretaking	$\beta \approx +0.22$	Care + protection predicts higher ratings
AX_payoff_safety	$\beta \approx +0.21$	Emotional safety / repair predicts higher ratings
AX_explicitness	$\beta \approx -0.15$	Tends negative once reach is controlled

AX_negative_affect	$\beta \approx -0.13$	Borderline negative baseline negativity
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Section F — Narrative arcs: timing matters

Instead of only averaging themes across a whole book, we look at **begin vs middle vs end**. Arc effects are differences like **end – begin**.

- Arcs capture **shape**: e.g., a late crisis vs early constant distress.
- In romance structure terms, this aligns with “third-act crisis then repair.”
- We treat arcs as planned measures (not conditional variants) to avoid p-hacking.

F2 — Arc effects plot + key coefficients

What we're seeing: some negative-affect themes increase from begin→end in higher-rated books. Why it matters: a **structured late crisis** can be part of a satisfying arc.

Slide 18: Narrative Arc / Pacing Results

Higher-Rated Books Show Better Pacing

Late-story crisis escalation (third-act crisis): - **F2_anger_frustration end-begin**: $\beta \approx +0.24$, CI [+0.08, +0.41], $P=0.995$ - **F3_anxiety_worry end-begin**: $\beta \approx +0.19$, CI [+0.02, +0.36], $P=0.981$

Interpretation

Higher-rated books have: - **Lower baseline negativity** across the book - **Stronger late "crisis escalation"** (third-act crisis) - Consistent with romance narrative structure (Radway phases)

Most Positive Arc Effects

1. **F2_anger_frustration end-begin**: $\beta \approx +0.24$, $P=0.995$
2. **F3_anxiety_worry end-begin**: $\beta \approx +0.19$, $P=0.981$
3. **O_aesthetics_appearance end-begin**: $\beta \approx +0.15$, $P=0.950$

Most Negative Arc Effects

1. **J_social_support_kin end-begin**: $\beta \approx -0.18$, $P=0.019$
2. **S_scene_anchors end-begin**: $\beta \approx -0.16$, $P=0.074$
3. **C_explicit_eroticism end-begin**: $\beta \approx -0.12$, $P=0.103$

Example: Radway Phase Distribution

Phase I: Initial Conflict & Isolation (147 topics, 54.0%) - Example topics: "Insane Arguments", "Relationship Ambiguity Conversation" - Radway functions: R2 (Heroine reacts antagonistically), R7 (Physical/emotional separation)

Phase II: Turning Point & Recognition (96 topics, 35.3%) - Example topics: "Tender Forehead Kisses", "Frightened Admissions" - Radway functions: R8 (Hero treats heroine tenderly), R9 (Heroine responds warmly)

Arc contrast (end-begin)	β (approx.)	Reading
Anger/Frustration (F2)	+0.24	Late crisis escalation
Anxiety/Worry (F3)	+0.19	Tension rises late then resolves
Explicit eroticism (C)	-0.12	Less likely to ramp late in higher-rated books

F3 — Plain-language arc takeaway + Radway phases

- Best interpretation is **not** “more misery is better.” It’s “timed conflict can make the payoff feel earned.”
- Higher-rated books show lower baseline negativity overall, but stronger late escalation (classic third-act crisis).
- Phase distribution example (topic mapping): Phase I 54.0%, Phase II 35.3%, Phase III 10.3%.

Radway phase	Share of mapped topics	Typical function
Phase I — Conflict & isolation	54.0% (147 topics)	Separation, antagonism, misunderstanding
Phase II — Recognition	35.3% (96 topics)	Tenderness, vulnerability, mutual warmth
Phase III — Restoration	10.3% (28 topics)	Commitment, declaration, HEA/repair

Section G — Predictive sanity check (cross-validation)

Cross-validation asks: if we predict unseen books, do themes add signal beyond metadata? With N=92, this is a reality check.

Slide 20: Predictive Performance

Cross-Validation Results (20 repeats of 5-fold CV)

Outcome	Model	CV_R ² _mean	CV_R ² _sd
rating_mean	metadata_only	0.108	0.031
rating_mean	metadata+core	0.056	0.041
log_rating_count	core_only	0.050	0.037

Interpretation

- Themes explain popularity (reach) better than star ratings
- Star ratings likely influenced by factors beyond theme indices (prose quality, pacing, editing, reader expectations)
- **Meta-result:** Market reach is more systematically related to thematic content than star ratings

Table: results/stage10_correlation_analysis/
04_hypothesis_testing/
inference_20260112_213754__measurement_v5/appendix_csv/
cv_results.csv

- Themes add **modest** out-of-sample signal for **Reach**.
- **Quality** is dominated by metadata/market factors at this sample size.
- Implication: the measurement works, but subtle quality effects likely need larger N and shrinkage models.

Wrap-up: What this pilot gives us (and what to do next)

Pilot bottom line: the instrument is working; larger N is the lever that will separate subtle quality effects from market reach.

Keep (locked-in):

- Normalized probability contracts (stop the pipeline if sums don't match).
- Auditable measurement bundle: topic membership + filter audits + CORE gating.
- Two-channel framing: Reach vs Quality.
- Macro axes for robust, communicable inference.

Add for the big corpus:

- Finer narrative bins → trajectory features (slope, peak position, curvature).
- Shrinkage models (ridge / Bayesian partial pooling) for many predictors.
- Bootstrap sign stability as the primary “trend” metric.

Generated on 2026-01-13.